

UDAY KIRAN CHERRI

CYBERSECURITY ENGINEERING - SECURITY MONITORING - DETECTION ENGINEERING

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LinkedIn - GitHub - Portfolio - Security Blog

SUMMARY

CS undergraduate focused on cybersecurity engineering, security monitoring, detection validation, and security research. Built reproducible security PoCs spanning intrusion correlation, runtime policy enforcement, telemetry validation, audit logging, and automated testing. Hands-on with Python, Bash, Docker, APIs, Prometheus, network-flow data, MITRE ATT&CK, and GitHub Actions.

SKILLS

- **Detection & Monitoring:** Security Monitoring, Detection Engineering, MITRE ATT&CK, Network-Flow Telemetry, Intrusion Correlation, Detection Validation, Audit Logging, Prometheus
- **Security Engineering:** Vulnerability Analysis, Runtime Policy Enforcement, Interprocedural Taint Analysis, SAST/SCA, Web & API Exploitation, Reverse Engineering
- **Languages & Scripting:** Python, Bash, Go, TypeScript, C++
- **Infrastructure & Automation:** Docker, Docker Compose, GitHub Actions, FastAPI, PostgreSQL, Kubernetes, Azure, GCP
- **Security Tooling:** Burp Suite, Nmap, Metasploit, Ghidra, IDA, GDB, pwntools
- **Security Research:** STIX 2.1, Federated Learning, Flower, FedProx, SecAgg+, PyTorch Geometric, Heterogeneous GNNs, GNNExplainer, SHAP

PROJECTS

FedShield-CTI - Federated Intrusion Detection

RGUKT Nuzvid Research Team

Python, PyTorch Geometric, Flower, STIX 2.1, MITRE ATT&CK

GitHub

- Built a heterogeneous CTI graph layer integrating STIX 2.1, MITRE ATT&CK, and network-flow data to correlate multi-source intrusion telemetry.
- Validated federated detection and privacy controls across three simulated organizations using Private Set Intersection and SecAgg+, reducing gradient-inversion reconstruction from 100% to 0%.
- Improved detection by +0.19 macro F1 over isolated local training and verified 20/20 federated rounds through an audit layer.

MCP Sentinel - Security Scanner & Runtime Guardrail

Independent Project

Python, FastAPI, Prometheus, GitHub Actions

GitHub

- Built an interprocedural taint-analysis engine tracing values across files and call chains to detect command injection, SQL injection, path traversal, and SSRF.
- Built a fail-closed runtime proxy for AI agents and MCP servers with policy-as-code, host/path/SQL containment, rate limiting, prompt-injection detection, and Prometheus metrics.
- Created 169 passing tests with CI-enforced SARIF scanning, SBOM generation, pip-audit, mypy, ruff, and an automated end-to-end attack validation path.

HT-Manager - CTF Team Operations Platform

HackerTroupe, RGUKT Nuzvid

Python, Discord.py, PostgreSQL, Docker, Azure

GitHub

- Built and deployed a production Discord bot automating event curation, workspace management, participation tracking, and result synchronization.
- Implemented layered architecture across 19 slash commands with audit logging, automated cleanup, CI gating, and reproducible Docker Compose deployment on Azure.

EXPERIENCE

CTF Lead & Security Operations Coordinator

May 2025 - Present

HackerTroupe, RGUKT Nuzvid

- Founded RGUKT Nuzvid's CTF team and grew it from 0 to 20+ members, reaching a top-13 national finish within the first year.
- Applied web exploitation, binary reverse engineering, cryptanalysis, and OSINT automation across 50+ competitions, with emphasis on repeatable technical investigation workflows.
- Built the team's website and cloud-hosted automation tooling, reducing competition setup time by 50%.

EDUCATION

B.Tech in Computer Science

2023 - May 2027

RGUKT Nuzvid

CTF HIGHLIGHTS & CERTIFICATIONS

- CTF: 2nd - CRACK-IN 1.0 (CyberHive, GGITS) • 13th - CyberTEA 3.0 (IIIT Sri City) • 37th - IIT Madras CTF • 98th / 843 - Idek CTF 2025 • 50+ competitions across web, binary, crypto, forensics, and OSINT.
- ISC2 Certified in Cybersecurity (CC) • JPMorgan Chase Cybersecurity Job Simulation • TryHackMe - Top 6% • Security Blue Team - Introduction to OSINT, Network Analysis, and Dark Web Operations.